

Shared rollout



$$\pi_{\theta} : x \mapsto \{y_1, \dots, y_G\}$$

focus response: y_i

$y_{i,1} \ y_{i,2} \ \dots \ y_{i,t} \ y_{i,T}$

$R_i : 0/1$ ℓ^T, ℓ^θ, H^T

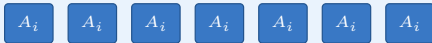


GRPO

$$A_{i,t}^{\text{GRPO}} = \frac{R_i - \mu_x}{\sigma_x + \epsilon}$$

μ_x, σ_x use peer responses; one scalar within y_i

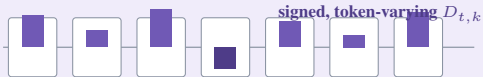
same advantage at every token



OPD

$$\delta_{t,k} = \ell_{t,k}^T - \ell_{t,k}^\theta, \quad D_{t,k} = \rho_{t,k} \delta_{t,k}$$

dense local signal, but no outcome or prefix validity check



POV-OPD: validate and calibrate the dense OPD signal

$$A_{t,k}^{\text{POV}} = D_{t,k} w_{t,k}^{\text{out}} w_t^{\text{pre}} w_{t,k}^{\text{sup}}$$



1. Outcome validity $\rightarrow w^{\text{out}}$

trajectory label: $z_i = 21[R_i > 1/2] - 1$

local alignment: $s_{t,k} = z_i \delta_{t,k}$

aligned: $s_{t,k} > m \rightarrow 1$

ambiguous: $|s_{t,k}| \leq m \rightarrow 0$

conflict: $s_{t,k} < -m \rightarrow \beta q_{t,k}$

$q = \text{rank}_{\mathcal{B}}(s)/(n_{\mathcal{B}} - 1)$; $q = 1$ for a singleton



2. Prefix validity $\rightarrow w^{\text{pre}}$

excess surprisal u_t

window mean $\bar{u}_b$

CUSUM drift C_b

first L windows $\rightarrow \mu$



$b < b^*$: full trust
 $w_b^{\text{pre}} = 1$

$b \geq b^*$: monotone decay
 $w_b^{\text{pre}} = \min(w_{b-1}^{\text{pre}}, e^{-\lambda(C_b - \tau)_+})$



3. Teacher support $\rightarrow w^{\text{sup}}$

$$p_{t,k}^{\text{sup}} = e^{-[-\ell_{t,k}^T - H_t^T]_+}$$

teacher plausibility controls the boost

$s > m$: $w^{\text{sup}} = 1 + \alpha p^{\text{sup}}$

otherwise: $w^{\text{sup}} = 1$

aligned-only boost; no sign flip

How the three weights change credit on the same response

$b_t \equiv 1 + \alpha p_t^{\text{sup}}$ for aligned tokens; $\gamma_6 \leq \gamma_5 < 1$ from drift onward

sampled-token example	t_1 aligned	t_2 aligned	t_3 ambiguous	t_4 conflicting	t_5 aligned + drift	t_6 aligned + later
raw OPD D_t	D_1	D_2	D_3	D_4	D_5	D_6
outcome w_t^{out}	1	1	0	βq_4	1	1
prefix w_t^{pre}	1	1	1	1	γ_5	γ_6
support w_t^{sup}	b_1	b_2	1	1	b_5	b_6
final A_t^{POV}	$D_1 b_1$	$D_2 b_2$	0	$\beta q_4 D_4$	$\gamma_5 D_5 b_5$	$\gamma_6 D_6 b_6$

Outcome calibrates OPD; it is not added as a separate GRPO reward and never flips D_t 's sign.